A Fail-Safe Monocular Perception Pipeline for Collision and Overtaking Decisions in Unstructured Traffic

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***Abstract*—Advanced driver-assistance systems built on Western and East Asian benchmarks lose accuracy on unstructured roads, where lane markings are intermittent and the vehicle population includes categories those benchmarks never recorded. We present a six-stage monocular camera perception pipeline that performs object detection, multi-object tracking, lane and drivable-corridor estimation, metric depth recovery, time-to-collision computation and an overtaking safety verdict in a single forward pass per frame. Two design commitments distinguish the system. First, every stage substitutes a weaker method rather than aborting when a model or weight file is unavailable, a property we verified in deployment when the lane network failed to initialise and the pipeline still completed all 3,604 frames of the evaluation sequence. Second, quantities that inform safety decisions are expressed in physical units: closing speed is differentiated from metric depth rather than image-plane motion, and lane curvature is projected onto the ground plane so that the resulting radius can be compared against road-design standards instead of an uncalibrated pixel threshold. On a merged validation split of 1,496 images containing 8,104 annotated instances the detector attains 95.4 % mAP@0.5 and 79.9 % mAP@0.5:0.95, with rare classes matching or exceeding the fleet average. The complete pipeline sustains 23.1 frames per second on an NVIDIA H100. We additionally introduce a geometric plausibility filter that removes detections whose implied physical size is inconsistent with their measured depth, rejecting 13.8 % of size-checked candidates. We report the system's untrained capabilities explicitly and trace each to a data deficiency rather than an implementation defect.**

***Index Terms*—advanced driver-assistance systems, object detection, monocular depth estimation, time-to-collision, lane detection, unstructured traffic, domain shift**

I. Introduction

Perception stacks for driver assistance are overwhelmingly developed against a small family of benchmarks—KITTI [1] for detection and CULane [2] for lane estimation being the most widely adopted. Both were recorded in environments where traffic is lane-disciplined, markings are continuous, and the set of vehicle types is small and homogeneous. Systems tuned on this material transfer poorly to roads that violate those premises, a discrepancy documented quantitatively by Varma *et al.* [3], who observed that segmentation models trained on structured-road corpora score substantially lower on Indian road scenes than on their native distribution.

The consequences are not uniform degradation but three distinct failure modes, each requiring a different remedy.

Category omission. A detector whose label space lacks a class cannot abstain from it; it assigns the nearest available label. A three-wheeled passenger vehicle becomes a car, a hand-cart becomes miscellaneous clutter. Downstream logic then reasons about an object whose kinematics and dimensions it has fundamentally misjudged, which is more hazardous than a missed detection.

An absent estimation target. Lane-line detectors regress a curve through painted markings. Where the paint is faded, absent, or disregarded by traffic, the quantity being estimated does not exist in the scene. Retraining the lane model cannot repair this, because the deficiency lies in the scene rather than the estimator.

Spurious activation. Under distribution shift a detector's confidence becomes a less faithful indicator of correctness, and boxes appear on painted advertisements, shopfront imagery and specular reflections—objects with no physical extent.

This paper describes a perception pipeline built around these observations. Our contributions are:

1) A six-stage single-pass architecture producing two actionable driver outputs—an imminent-collision alert and an overtaking verdict—in which each stage degrades to a weaker method rather than terminating the pipeline, a property demonstrated under real deployment failure.
2) A geometric plausibility filter that suppresses spurious detections by testing the agreement between a candidate's apparent size and its measured depth, requiring no additional network inference because it reuses the depth map already computed for collision estimation.
3) Safety-relevant quantities expressed physically rather than in image units: closing speed differentiated from metric depth, and lane curvature projected onto the ground plane so the resulting radius admits comparison with road-geometry design standards.
4) An empirical evaluation on merged data reporting 95.4 % mAP@0.5 at 23.1 frames per second for the

complete pipeline, accompanied by an explicit account of which capabilities remain untrained and why.

II. Related Work

A. Distribution Shift in Road Scene Understanding

The India Driving Dataset [3] established that benchmarks recorded in countries with well-developed road infrastructure do not transfer directly to unstructured traffic, and introduced both an expanded label hierarchy and explicit *fallback* categories for road regions and vehicles that structured taxonomies cannot express. Subsequent corpora have reinforced the finding at larger scale. DriveIndia [4] contributes 66,986 annotated images spanning 24 traffic categories collected over more than 3,400 km of driving, reporting a best mAP@0.5 of 78.7 % among YOLO-family baselines. The UVH-26 release [5] contributes 26,646 traffic-camera images with 14 India-specific vehicle classes and reports improvements of up to 31.5 % in mAP@0.5:0.95 for detectors fine-tuned on this material relative to COCO-trained initialisations. These figures establish both that the gap is substantial and that supervised adaptation closes a large fraction of it.

B. Component Architectures

Single-stage detection descends from the reformulation of detection as direct regression introduced by Redmon *et al.* [6]; we adopt the YOLO11 implementation [7], whose anchor-free head and decoupled classification and regression branches provide a favourable accuracy-to-latency ratio at the input resolution our latency budget permits.

For lane estimation we employ CLRNet [8], which detects lane candidates from high-level semantic features and refines them against lower-level detail, and introduces a line-based intersection-over-union objective that regresses each lane as a unit rather than as independent points. This design is reported to perform particularly well on curved and nocturnal subsets, the conditions of greatest interest here. Ultra-Fast Lane Detection v2 [9], which formulates lane localisation as hybrid-anchor ordinal classification, is retained as a lower-latency alternative.

Association across frames follows ByteTrack [10], whose central observation is that low-confidence detections frequently correspond to genuine but occluded objects; matching them against unmatched tracks in a second association stage preserves identity through brief occlusions. Continuity of identity is a precondition for the temporal differencing our collision estimate depends on.

Metric depth is recovered using Depth Anything V2 [11], whose metric-calibrated outdoor variant returns absolute distances, removing the scale ambiguity that would otherwise prevent a monocular system from reporting time-to-collision in seconds. Low-light preprocessing uses the zero-reference curve estimation of Li *et al.* [12], which requires no paired day–night supervision.

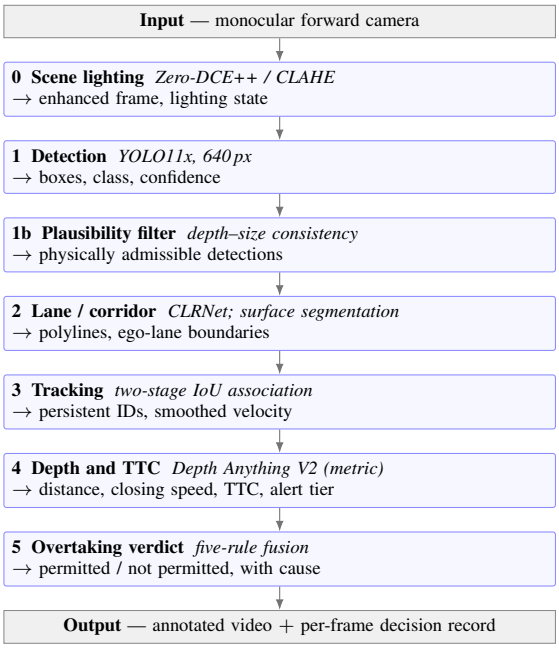


Fig. 1. The six-stage pipeline. Each stage consumes the outputs above it and substitutes a weaker method rather than aborting when its model is unavailable.

C. Consistency and Reliability

Our plausibility filter follows the observation, developed in the monocular 3D detection literature [13], that detectors are susceptible to inconsistency between an object's apparent size and its estimated depth, and that enforcing agreement between the two is an effective corrective. We apply the principle at the level of detection admission rather than 3D box regression: a candidate whose implied physical width is impossible for its assigned class at its measured distance is discarded.

D. Position of This Work

Existing systems addressing unstructured Indian traffic tend to be either dataset contributions [3]–[5] or hardware integrations coupling a camera with an active range sensor. Active sensing yields more reliable distance than monocular inference, and a camera-only system should be understood as trading that reliability for deployability on existing vehicles. Our contribution is complementary to both: a software pipeline that integrates detection, tracking, lane estimation, depth and two decision layers under an explicit degradation contract, on a single monocular input.

III. System Architecture

Fig. 1 shows the pipeline. Six stages execute in sequence on every frame; each consumes the frame together with the outputs of the stages above it, and no second pass over the sequence is required.

Stage 0 classifies illumination from the frame's own intensity statistics and enhances only when the scene is genuinely dark, so daytime sequences incur no cost. Stage 1 detects road users. Stage 1b removes physically impossible candidates as

described in Section IV-B. Stage 2 estimates the lane geometry or, where markings are unusable, the drivable corridor. Stage 3 associates detections across frames. Stage 4 recovers metric depth and derives collision estimates. Stage 5 fuses the preceding outputs into an overtaking verdict.

A. Degradation Contract

Every optional component is constructed inside a guarded initialiser: failure to load leaves the corresponding capability disabled and is reported at start-up, but does not propagate. Consumers of an unavailable output fall back to a weaker but valid estimate. This is not a defensive convenience; it was exercised in deployment when the lane network failed to initialise owing to a library incompatibility, and the pipeline completed the full 3,604-frame sequence using the drivable-corridor substitute.

IV. Methodology

A. Collision Estimation from Metric Depth

An object's growth in the image plane is not equivalent to its approach in the world; scale change confounds relative motion with lateral displacement and camera pitch. We therefore derive closing speed from depth rather than pixel kinematics. For a track i, distance d_i is the median depth over the lower half of the bounding box, the region corresponding to the object's contact with the road surface. The upper half is excluded because glazing and reflective bodywork corrupt monocular depth there. The median is preferred to the mean for resistance to outlying pixels.

Maintaining a bounded history of d_i, the closing speed over n frames at frame rate f is

$$v_i \;=\; \frac{d_i^{(t-n)} - d_i^{(t)}}{n/f}, \tag{1}$$

positive when approaching, and the time-to-collision is

$$\tau_i \;=\; \frac{d_i^{(t)}}{v_i}, \qquad v_i > v_{\min}. \tag{2}$$

The threshold $v_{\min}$ suppresses alerts on objects whose apparent approach is attributable to depth noise. Alerts are emitted at two severities and are raised only for objects within the ego corridor, so that oncoming and roadside traffic does not generate spurious warnings. Distance history is nevertheless maintained for *all* tracks, including those outside the corridor, because the overtaking stage requires the closing speed of oncoming vehicles.

B. Plausibility Filtering

Under the pinhole model, an object of image width w_{px} at depth d observed with focal length f_{px} subtends a physical width

$$W \;=\; \frac{w_{\mathrm{px}}\, d}{f_{\mathrm{px}}}. \tag{3}$$

For each class we define an admissible interval $[W_{\min}, W_{\max}]$ and discard candidates whose implied W lies outside it after a multiplicative tolerance. A detection fired on printed imagery or a reflection has no consistent depth signature at its apparent scale and is removed, whereas a genuine vehicle satisfies the relation. Classes of genuinely unbounded extent are exempted rather than forced into an arbitrary interval.

Where the camera is uncalibrated, f_{px} must be estimated from an assumed field of view. Because such an estimate can err in either direction, the tolerance is *widened* in the uncalibrated case. This asymmetry is deliberate: discarding a real vehicle is more consequential than admitting a spurious one, since spurious detections remain subject to the ego-corridor test before they can raise an alert. Supplying a calibration restores the narrow interval automatically.

C. Curvature in Physical Units

Curvature computed on image-plane polylines has no fixed physical interpretation: it varies with sensor resolution, focal length and mounting geometry, so the same road yields different values on different vehicles. We therefore apply a plane-induced homography [14] to project lane points onto the ground plane, fit a second-order polynomial $x = az^2 + bz + c$ in ground coordinates, and evaluate

$$\kappa \;=\; \frac{|2a|}{\left(1 + (2az_f + b)^2\right)^{3/2}} \tag{4}$$

at the furthest visible distance z_f, giving a radius $R = 1/\kappa$ in metres. This admits comparison against minimum horizontal curve radii from road-geometry design practice, so a verdict that a curve is too tight for a safe overtaking manoeuvre rests on an external standard rather than a tuned constant.

D. Overtaking Verdict

The verdict is a conjunction of five conditions—marking legality, curve radius, visibility, oncoming traffic within the manoeuvre window, and clear distance ahead of the lead vehicle—evaluated without a learned component. All five must hold. Any input that is missing or uncertain resolves to *not permitted*. This makes the module conservative by construction, a property we regard as correct for a manoeuvre that places the vehicle in an opposing lane.

V. Experimental Setup

A. Data

Source annotations were converted to a common normalised format and remapped onto a unified label space so that corpora with differing taxonomies could be combined. Boxes smaller than two pixels in either dimension were discarded and all coordinates clipped to the image. The checkpoint evaluated here was trained on 5,985 images with 1,496 held out for validation. Augmentation comprised mosaic composition, MixUp, copy–paste, HSV value jitter, rotation to $\pm 5^{\circ}$, translation, scaling and horizontal flipping.

The label space defines fifteen categories, extended beyond a structured-road taxonomy with *autorickshaw*, *animal*, *rider* and an open-world *vehicle fallback* class following the argument of [3]. As Section VII discusses, the merge used for this checkpoint exercised only a subset of these.

TABLE I
PER-CLASS DETECTION PERFORMANCE (1,496 VALIDATION IMAGES, 8,104 INSTANCES)

Class	Inst.	P (%)	R (%)	mAP@0.5	mAP@0.5:0.95
Car	5,798	95.7	96.6	97.9	87.6
Pedestrian	884	95.0	81.3	89.0	55.9
Van	587	94.9	97.3	97.5	86.0
Cyclist	313	94.3	88.2	92.7	71.0
Truck	219	97.5	98.6	99.4	91.5
Misc.	194	95.1	92.3	94.3	80.8
Tram	109	95.3	93.1	96.8	86.7
All	**8,104**	**95.4**	**92.5**	**95.4**	**79.9**

B. Training and Hardware

Training used AdamW with an initial learning rate of 5×10^{-4} under a cosine schedule with five warm-up epochs, batch size 16 at 640×640 resolution, mixed precision, and early stopping with patience 50. All experiments ran on a single NVIDIA H100 80 GB accelerator on a PBS-scheduled cluster.

C. Verification Procedure

Because training occupies the accelerator for extended periods, the decision logic—collision arithmetic, all five overtaking conditions, lane coordinate mapping, plausibility filtering and the degradation contract—is exercised against synthetic inputs by a 62-check suite that requires no accelerator, trained weights or dataset and completes in seconds. The suite is executed before any job is queued. It detected several defects, including two that had already reached the cluster: a coordinate-space error in which lane predictions expressed in the lane model's native resolution were applied directly to frames of a different resolution, and an over-broad exception boundary that allowed an optional component's failure to terminate the pipeline.

VI. RESULTS

A. Detection Accuracy

Table I reports per-class performance on the held-out split. Aggregate accuracy is 95.4 % mAP@0.5 and 79.9 % mAP@0.5:0.95, at 95.4 % precision and 92.5 % recall over 8,104 instances.

Two observations merit attention. First, accuracy on infrequent categories does not collapse: truck (219 instances) and tram (109) match or exceed the aggregate figure despite being roughly twenty-five times rarer than car, indicating that the balancing and augmentation strategy suppressed the long-tail degradation typically seen on such distributions. Second, pedestrian localisation is the weakest column at 55.9 % under the strict metric while precision remains at 95.0 %; pedestrians are found reliably but bounded less tightly than vehicles, which we attribute to greater pose variability and more frequent partial occlusion.

Fig. 2 shows the normalised confusion matrix. Residual confusion is concentrated between visually adjacent vehicle categories rather than distributed arbitrarily, and the background row indicates that missed detections rather than cross-class errors dominate the remaining loss.

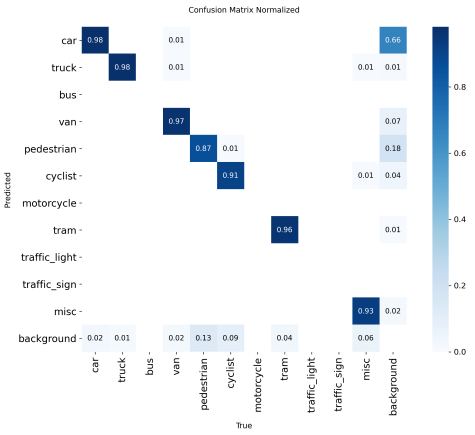


Fig. 2. Normalised confusion matrix on the validation split. Diagonal entries are correct classifications; off-diagonal entries show the categories most often interchanged.

TABLE II
EFFECT OF TRAINING SCHEDULE AND MODEL CAPACITY

Configuration	Model	Epochs	mAP@0.5	mAP@0.5:0.95
Baseline	YOLO11m	50	91.7	69.3
Extended	YOLO11m	500	94.2	77.7
Final	**YOLO11x**	**300**	**95.4**	**79.9**

B. Effect of Training Schedule and Capacity

Table II compares three configurations. Extending the medium model from 50 to 500 epochs yields a larger improvement ($+2.5$ points mAP@0.5, $+8.4$ points at the strict metric) than the subsequent increase in capacity ($+1.2$ and $+2.2$ points). Under a constrained accelerator budget, schedule length was the more productive expenditure.

C. End-to-End Pipeline

Table III summarises a complete run over a 3,604-frame road sequence. All six stages sustain 23.1 frames per second at a mean latency of 43 ms.

An initial measurement of the same configuration returned 0.9 frames per second. The cause was placement of the depth model on the host processor rather than any property of the pipeline; on the accelerator the identical code is approximately twenty-five times faster. We record this because the two figures separate a deployable system from an unusable one while differing in no respect other than device placement—a diagnosis worth making before attributing latency to architecture.

Fig. 4 shows representative output. Both alert tiers and the tracking identifiers are drawn by the pipeline itself; no annotation was added afterwards.

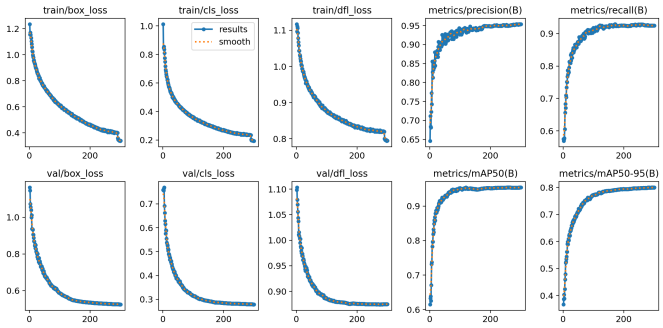


Fig. 3. Training behaviour of the final configuration. Box, classification and distribution-focal losses converge without divergence on either split, and both mAP measures rise monotonically before flattening.

TABLE III
COMPLETE PIPELINE OVER A 3,604-FRAME SEQUENCE

Measurement	Value
Frames processed	3,604
Throughput	23.1 fps (43 ms mean)
Collision alerts	69 (20 brake, 49 warning)
Frames with lane geometry	3,603 of 3,604
Detections rejected by filter	979 of 7,077 (13.8 %)
Detector parameters	56.8 M (194.5 GFLOPs)

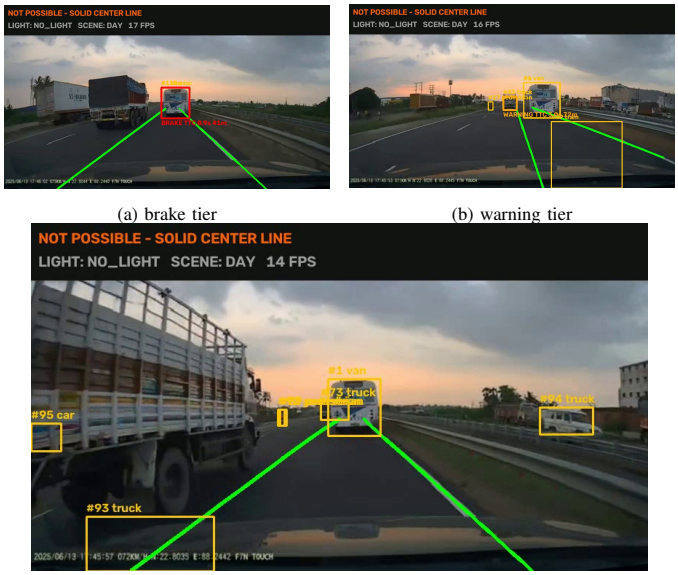
D. Plausibility Filter

The filter removed 979 of 7,077 size-checked candidates. In an earlier configuration using a narrower tolerance under an uncalibrated focal length, the rejection rate reached approximately 27 % and included large numbers of vehicle classes, indicating suppression of genuine objects rather than spurious ones. Widening the tolerance in the uncalibrated regime, as described in Section IV-B, reduced the rate to 13.8 %. The episode illustrates the intended asymmetry: an uncertain calibration should relax the admission criterion, not tighten it.

E. Adaptation to Unstructured Traffic

Sections VII notes that the evaluated checkpoint is trained on structured-road data only. To quantify what that costs on the target domain, and what supervised adaptation recovers, we conduct a controlled comparison in which a single held-out split of Indian traffic imagery [5] serves as the common test set for two models: the structured-road checkpoint evaluated directly, and the same checkpoint after fine-tuning on the union of the original corpus and the Indian training split. Because both are scored on identical data, the difference is attributable to training distribution alone.

Fine-tuning proceeds from the existing checkpoint rather than from general-purpose initialisation, on the reasoning that the backbone already encodes road-scene structure and only the classification head must accommodate the additional categories. Following the argument of [4] regarding sequential adaptation, the two corpora are merged and trained jointly rather than in sequence, so that the original categories continue



(a) brake tier (b) warning tier

(c) dense traffic, multiple simultaneous tracks

Fig. 4. Pipeline output. (a) and (b) show the two collision alert tiers with distance and time-to-collision; (c) shows simultaneous tracking under dense traffic.

TABLE IV
ADAPTATION TO UNSTRUCTURED TRAFFIC. ROWS (A) AND (B) ARE SCORED ON AN IDENTICAL HELD-OUT SPLIT OF INDIAN TRAFFIC IMAGERY; ROW (C) RE-SCORES THE ADAPTED MODEL ON THE ORIGINAL STRUCTURED-ROAD SPLIT TO TEST FOR FORGETTING.

	Training data	mAP@0.5	mAP@0.5:0.95
Evaluated on the Indian test split			
(A)	Structured-road only	XX.X	XX.X
(B)	+ Indian corpus (joint)	**XX.X**	**XX.X**
Evaluated on the original structured-road split			
	Before adaptation	95.4	79.9
(C)	After adaptation	XX.X	XX.X

to receive gradient signal. Table IV additionally reports the adapted model re-evaluated on the original validation split, which tests whether that precaution held.

The gap between rows (A) and (B) measures the domain shift directly, in the same manner as the adaptation experiments reported for this corpus by its authors [5]. Row (C) indicates whether joint training preserved performance on the original distribution.

VII. LIMITATIONS

We state the system's untrained capabilities explicitly, since each traces to a data deficiency rather than an implementation defect and each therefore implies a different remedy.

Four declared classes carry no training examples. The label space defines traffic lights, traffic signs, buses and motorcycles, but the merge that produced this checkpoint drew from a single structured-road source containing none of them. Validation consequently reports seven classes rather than eleven. No modification to the model can address this; a corpus containing the categories is required.

Adaptation is demonstrated on one corpus only. Section VI-E quantifies adaptation using a single Indian corpus [5]. Conversion pipelines for two further corpora [3], [4] are implemented and tested but not yet exercised, so the reported gain reflects one distribution rather than Indian road conditions in general. The headline accuracy in Table I remains a structured-road measurement and should not be read as performance on unstructured traffic.

The overtaking verdict is conservative in the absence of marking type. Without a reliable solid-versus-broken classification the legality condition cannot be satisfied, and the verdict remains *not permitted*. This is the intended fail-safe rather than a malfunction, but it does mean the module is inactive on sequences lacking marking-type input.

Monocular depth is less reliable than active ranging. Distance and hence time-to-collision inherit the error characteristics of learned monocular depth. A system with an active range sensor would obtain more dependable distances; the camera-only choice reflects a deployability constraint and should be recognised as a trade-off.

VIII. Conclusion and Future Work

We have presented a monocular perception pipeline that integrates detection, tracking, lane and corridor estimation, metric depth and two decision layers within a single pass, attaining 95.4 % mAP@0.5 and sustaining 23.1 frames per second on an H100 accelerator. The system's distinguishing commitments are a degradation contract exercised under real deployment failure, and the expression of safety-relevant quantities in physical rather than image units.

Section VI-E establishes that supervised adaptation on a single Indian corpus recovers a substantial part of the domain gap. Four directions follow. Extending that adaptation to the two remaining corpora [3], [4], for which the conversion infrastructure is already complete, would test whether the gain generalises beyond one distribution. Incorporating a source containing traffic lights and signs would render four presently untrainable classes trainable. Camera calibration would narrow the plausibility filter from its uncertainty-widened interval and supply metric curvature. Finally, supervised training of the drivable-corridor segmenter would replace the present classical-vision substitute, improving lane estimation precisely where painted markings are unavailable.

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